

Experiment planning

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We will consider some questions
related to the description of experimental results
and the planning of experimental conditions.

Chapter 1. Distribution laws of random variables

1.1. Distribution of random discrete variables

Random variables can be discrete or continuous.

Consider random discrete variables.

How to plot a distribution for discrete variables?

Example 1.1

Let us have a collection of 8 groups of 3 balls.

RRR	RRB	RBR	BRR
RBB	BRB	BBR	BBB

X number of red balls in the group.

P(X) is a probability of finding a group with the number of red balls equal to X

P (X) is equal to the number of groups containing X red balls divided by the number of all groups

$$P(X) = \frac{\text{the number of groups containing X red balls}}{\text{the number of all groups}}$$

Example 1.1

(continuation)

RRR	RRB	RBR	BRR
RBB	BRB	BBR	BBB

$$P(X=0) = 1/8$$

$$P(X=1) = 3/8$$

$$P(X=2) = 3/8$$

$$P(X=3) = 1/8$$

We got the distribution of a random variable X

X	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

An important relationship for the probabilities of random variables:

The sum of all probabilities is one

$$\sum_{i=1}^{All} P(X_i) = 1$$

How we plot this, to see how this is distributed



Vertical axis for probabilities.
The probability value does not exceed one and is a multiple of $1/8$.
Marking half.
Marking half.
Marking half.
In this way, we get 8 marks on the vertical axis.

Horizontal axis for X
X could be zero
X could be one
X could be two
X could be three

We have made a probability distribution for the random variable X.
And the random variable X can only take on these discrete values.
It can't take on the value $5/8$ or the value $7/8$ or anything like that.

Example 1.2

Let us have a collection of 8 groups of 3 balls.

RRR	RRB	RBR	BRR
RBB	BRB	BBR	BBB

Y is number of blue balls in the group.
Construct a distribution for a random variable Y.

1.2 Checking the discrete probability distribution

Example 1.3

The farm has 15% white cows,
36% dark,
47% brown.

White	0.15
Dark	0.36
Brown	0.47

Is this probability distribution correct?

Recall that the sum of all probabilities must equal one.

$$\sum_{i=1}^{All} P(X_i) = 1$$

Calculate the sum of probabilities
 $0.15 + 0.36 + 0.47 = 0.98$

This is less than one and this distribution is not valid.

Example 1.4

The basket contains 33% red apples,
18% green,
62% yellow.

Is this probability distribution correct?

1.3 Calculation of the probability of a discrete random variable

Example 1.5

Let an incomplete probability distribution of a random variable be given.

Find $P(X=5)$

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	?

Recall that the sum of all probabilities must equal one.

$$\sum_{i=1}^{All} P(X_i) = 1$$

Calculate the sum of probabilities
 $0.21 + 0.13 + 0.24 + 0.42 = 0.90$

$$P(X=5) = 1 - 0.90 = 0.10$$

Thus, we have obtained the following probability distribution of the random variable X

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

Find the probability of X greater than two.

$$P(X>2) = P(3) + P(4) + P(5) = 0.24 + 0.32 + 0.10 = 0.66$$

$$**P(X>2) = 0.66**$$

Example 1.6

Let an incomplete probability distribution of a random variable be given

X	0	1	2	3	4
P(X)	0.05	0.31	0.15	?	0.27

Find $P(X=3)$

Find the probability of X greater than two.

1.4 Mean (expected value) of a discrete random variable

The mean or expected value $E(X) = \mu_X$

of a random discrete variable can
be calculated as follows

$$E(x) = \mu_X = \sum_{i=1}^{All} X_i \cdot P(X_i)$$

$$E(x) = \mu_X = \sum_{i=1}^{All} X_i \cdot P(X_i)$$

Example 1.7

Let an probability distribution of a random variable be given
Find mean or expected value

$$E(X) = \mu_X$$

X	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

$$E(X) = \mu_X = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8}$$

$$E(X) = \mu_X = 0 \cdot 0.125 + 1 \cdot 0.375 + 2 \cdot 0.375 + 3 \cdot 0.125 = 0 + 0.375 + 0.750 + 0.375 = 1.5$$

Thus, we got the expected value
and

$$E(X) = \mu_X = 1.5$$

this is the center of the distribution for a random variable X

Example 1.8

Let an probability distribution of a random variable be given

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

Find mean or expected value $E(X) = \mu_X$

1.5 Dispersion or variance and standard deviation of a discrete random variable

Example 1.9

Consider the probability distribution of a random discrete variable from **Example 1.7**

Mean or expected value $E(X) = \mu_X = 1.5$ sets the distribution center.

X	0	1	2	3
P(X)	0.125	0.375	0.375	0.125

In addition, variance (dispersion) and standard deviation are important characteristics of the distribution

The variance can be calculated by the formula

$$D(X) = E(X - \mu_X)^2 = \sum_{i=1}^{All} (X_i - \mu_X)^2 \cdot P(X_i)$$

Standard deviation

$$\sigma = \sqrt{D(X)}$$

Let's calculate the variance
in our example

$$E(X) = \mu_X = 1.5$$

X	0	1	2	3
P(X)	0.125	0.375	0.375	0.125

$$D(X) = (0 - 1.5)^2 \cdot 0.125 + (1 - 1.5)^2 \cdot 0.375 + (2 - 1.5)^2 \cdot 0.375 + (3 - 1.5)^2 \cdot 0.125$$

$$D(X) = 1.5^2 \cdot 0.125 + 0.5^2 \cdot 0.375 + 0.5^2 \cdot 0.375 + 1.5^2 \cdot 0.125$$

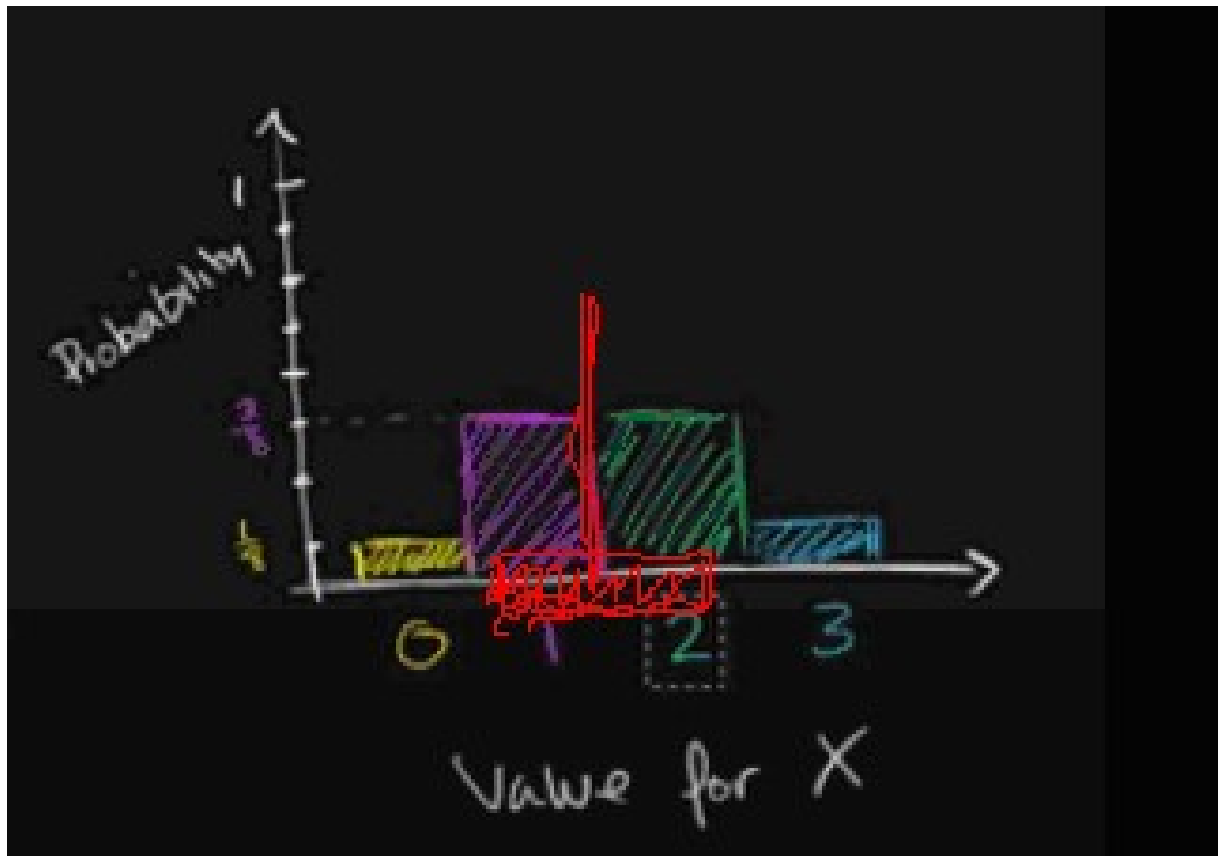
$$D(X) = 2.25 \cdot 0.125 + 0.25 \cdot 0.375 + 0.25 \cdot 0.375 + 2.25 \cdot 0.125$$

The variance is

$$D(X) = 0.75$$

Standard deviation

$$\sigma = \sqrt{D(X)} = \sqrt{0.75} = 0.866$$



$$E(X) = \mu_X = 1.5$$

Mean is a vertical red line

$$\sigma = \sqrt{D(X)} = 0.866$$

The standard deviation is the horizontal red line

Example 1.10

Consider the probability distribution of a random discrete variable from **Example 1.2**

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

Example 1.11

Consider the probability distribution of a random discrete variable from **Example 1.8**.

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

Find variance (dispersion) and standard deviation.